A　本文所用全部测试集内容

为了较为全面地考察大语言模型在化学核心任务上的推理表现，本研究自行设计并整理出了一套规模在 700 道题以上的综合化学测试集。该测试集在覆盖面上较为宽泛，既涉及从简单无机物到结构复杂的分子等不同类型的化学物质，也兼顾了从一维字符串到三维晶体结构等多种表达层级。为便于后续工作的复现与参考，下文将本文九项核心任务所对应的全部测试内容统一汇总于此：

A.1　分子名称与 SMILES 代码的双向转换测试集

Methane；Ethane；Propane；Butane；Ethene；Propene；Acetylene；Propyne；Methanol；Ethanol；Formaldehyde；Acetaldehyde；Formic acid；Acetic acid；Propionic acid；Acetone；Diethyl ether；Methylamine；Ethylamine；Acetonitrile；Dimethylamine；Isobutane；Neopentane；Isopropanol；Tert-Butanol；Triethylamine；Chloromethane；Dichloromethane；Chloroform；Carbon tetrachloride；Methyl tert-butyl ether；Benzene；Naphthalene；Anthracene；Phenanthrene；Toluene；Ethylbenzene；Phenol；Aniline；Nitrobenzene；Benzoic acid；Benzaldehyde；Cyclopropane；Cyclobutane；Cyclopentane；Cyclohexane；Cyclohexene；Pyridine；Pyridazine；Pyrimidine；Pyrazine；Pyrrole；Imidazole；Pyrazole；Furan；Thiophene；Oxazole；Isoxazole；Thiazole；Piperidine；Morpholine；Ethylene glycol；Glycerol；Urea；(Z)-But-2-ene；(E)-But-2-ene；(Z)-1,2-Dichloroethene；(E)-1,2-Dichloroethene；Maleic acid；Fumaric acid；(R)-Butan-2-ol；(S)-Butan-2-ol；(R)-Limonene；(S)-Limonene；(R)-Ibuprofen；(S)-Ibuprofen；(R)-Thalidomide；(S)-Thalidomide；(+)-Tartaric acid；(-)-Tartaric acid；*meso*-Tartaric acid；D-alanine；L-alanine；L-Leucine；D-Leucine；L-Cysteine；D-Cysteine；L-Fructose；D-Fructose；L-Malic acid；D-Malic acid；L-Glyceraldehyde；D-Glyceraldehyde；Cholesterol；Testosterone；Estradiol；Progesterone；Dexamethasone；Cortisone；Caffeine；Nicotine；Morphine；Codeine；Heroin；Quinine；Aspirin；Penicillin G；Doxorubicin；Vincristine；Paclitaxel；Erythromycin A；Streptomycin；Tetracycline；Tetrodotoxin；Saxitoxin；Sorafenib；Imatinib；Adamantane；Norbornane；Bicyclo[2.2.2]octane；Camphor；Borneol；alpha-Pinene；Spiro[4.5]decane；Spiro[2.2]pentane；Spiro[3.3]heptane；1,3-Cyclohexanedione；Fullerene C60；18-Crown-6；[2.2.2]Cryptand；Sodium acetate；Ammonium chloride；Copper(II) sulfate；Silver nitrate；Ethylmagnesium bromide；Tetrabutylammonium hydroxide；Deuterium oxide；Deuterated acetone；[^{13}C]Methane；Tritiated benzene；[^{15}N]Glycine；[^{14}C]Acetic acid (carboxy-labeled)；[^{14}C]Acetic acid (methyl-labeled)；Phenanthrene-D10；p-Terphenyl-d14 以及上述分子对应的 SMILES 字符串。

A.2　辛醇-水分配系数（log P）预测测试集

Methane；Ethane；Propane；Butane；Pentane；Hexane；Heptane；Octane；Methanol；Ethanol；1-Propanol；1-Butanol；1-Pentanol；1-Hexanol；Formic acid；Acetic acid；Propanoic acid；Butyric acid；Benzene；Toluene；Ethylbenzene；Naphthalene；Biphenyl；Anthracene；Acetone；Butanone；Acetaldehyde；Acetophenone；Diethyl ether；Tetrahydrofuran；Anisole；Methylamine；Triethylamine；Aniline；Chloroform；Carbon tetrachloride；Chlorobenzene；Bromobenzene；Iodobenzene；Pyridine；Ethyl acetate；Phenol；Nitrobenzene；Cyclohexane；Glycine；Alanine；Valine；Phenylalanine；Tryptophan；Aspartic Acid；Lysine；Ibuprofen；Aspirin；Sertraline；Fluoxetine；Atenolol；Metformin；Indigo；

beta-Carotene；linalool；limonene；gamma-Terpinene；(+)-Camphene；camphor；eugenol；citronellal；citronellol；p-cymene；(R)-(-)-Carvone；(1R)-(-)-fenchone；geraniol；carvacrol；thymol；alpha-terpinene；(1R)-Camphor；Eucalyptol；L-(-)-menthol；(-)-menthone；(-)-*cis*-Myrtanylamine；(1S)-(-)-Verbenone；Linalool oxide；Piperitone；1,4-Cineole；Myrtenal；(-)-Borneol；Dihydrocarvone；(+)-Pulegone；(-)-Carveol；(-)-Perilla aldehyde；(S)-*cis*-Verbenol；(+)-Fenchol；(+)-Isomenthol；Limonene oxide；Dihydrocarveol；Terpinen-4-ol；alpha-Terpineol；Methyleugenol；Nerol；beta-lonone；Linalyl acetate；Neryl acetate；Menthyl acetate；Geranyl acetate；alpha-Terpinolene；Car-3-ene；(+)-alpha-Pinene

A.3 极性表面积（PSA）预测测试集

diazepam；oxazepam；temazepam；lorazepam；nitrazepam；clonazepam；flurazepam；flutoprazepam；estazolam；midazolam；zopiclone；zolpidem hemitartrate；ramelteon；barbituric acid；Amobarbital；pentobarbital；phenobarbital；secobarbital；hexobarbital；phenytoin；ethotoin；fosphenytoin；trimethadione；dimethadione；phensuximide；methsuximide；ethosuximide；carbamazepine；oxcarbazepine；Vigabatrin；gabapentin；tiagabine；sodium valproate；zonisamide；topiramate；lamotrigine；chlorpromazine；acetopromazine；perphenazine；fluphenazine；fluphenazine enanthate；phenazine decanoate；thioridazine；thiethylperazine；pipothiazine；trifluoperazine；chlorprothixene；zuclopenthixol；flupenthixol；thiotixene；haloperidol；trifluperidol；spiperone；droperidol；timiperone；penfluridol；fluspirilene；pimozide；metoclopramide；sulpiride；remoxipride；nemonapride；tiapride；clozapine；clothiapine；olanzapine；quetiapine；risperidone；aripiprazole；loxapine；amoxapine；mosapramine；ziprasidone；moclobemide；reboxetine；amoxapine；maprotiline；desipramine；clomipramine；trimipramine；doxepin；amitriptyline；nortriptyline；fluoxetine；paroxetine；citalopram；sertraline；mirtazapine；venlafaxine；desvenlafaxine；duloxetine；milnacipran；bupropion；buspirone；chlormezanone；tandospirone；lithium carbonate；bromocriptine；apomorphine；pramipexole；ropinirole；talipexole；levodopa；carbidopa；benserazide；selegiline；rasagiline；safinamide；entacapone；tolcapone；nebicapone；memantine；amantadine；riluzole；tacrine；donepezil；rivastigmine tartrate；galanthamine；morphine；codeine；heterocodeine；thebaine；etorphine；naloxone；naltrexone；levorphanol；butorphanol；pentazocine；pethidine；anileridine；fentanyl；alfentanil；sufentanil；carfentanil；remifentanil；methadone；dextromoramide；dextropropoxyphene

A.4 分子几何构型判定测试集

BeCl2；CH4；PCl5；SF6；CO2；BF3；SiH4；AsF5；XeF6；CCl4；SbF5；NH3；H2O；SO2；SF4；ClF3；XeF2；PH3；H2S；BrF5；XeF4；OF2；PCl3；BrF3；XeO3；ICl3；NO3-；SO42-；PO43-；ClO3-；CO32-；NH4+；I3-；ICl4-；O3；NO2-；N3-；SO3；SCN-；AsH3；H2Se；KrF2；TeCl4；TlCl3

A.5 分子点群对称性信息识别测试集

2,3-Pentadiene；Triphenylphosphine oxide；H2O；SCl4；BrF3；Cyclopentadiene；4-chloropyridine；1,5-Naphthyridine；1,4-Cyclohexanedione；1,3,5-Trioxane；Isobutane；NH3；XeOF5-；Acetylene；Hydrogen Cyanide；Quinoline；N2O3；*meso*-Tartaric Acid；4,4'-Dichlorobiphenyl；Twistane；1,4-benzoquinone；Naphthalene；Ethene；AsF5；

PF5；Fe(CO)5；Cyclobutane；Tetraphenylene；Chair conformation of cyclohexane；S8；Staggered Ferrocene；Eclipsed Ferrocene；Benzene；Uranocene；Carbon dioxide；Oxygen；[Co(NO2)6]3-；P4；Carbon tetrachloride；Adamantane；CH4；SF6；1,3,5,7-Cyclooctatetraene；C60

A.6 ^{1}H NMR 谱图信号数量预测测试集

Methane；Ethane；Propane；n-Butane；Isobutane；Neopentane；Cyclopropane；Cyclobutane；Cyclopentane；Cyclohexane；Chloromethane；Bromoethane；1-Propanol；Isopropanol；Diethyl ether；Ethyl methyl ether；Acetone；Acetaldehyde；Acetic acid；Methyl acetate；Ethylamine；1-Butene；1-Butyne；Benzene；Toluene；tert-Butanol；Isopropylamine；Vinyl chloride；Propene；Lactic acid；Acrylic acid；3-Hydroxypropionaldehyde；Ethyl acetoacetate；1,2-Dichloroethane；1,1-Dichloroethane；Ethylene glycol；p-Xylene；o-Xylene；m-Xylene；Acetophenone；Benzoic acid；Benzaldehyde；Cinnamic acid；4-Nitrotoluene；2-Nitrotoluene；3-Nitrotoluene；Aniline；N,N-Dimethylformamide；Furfural；1,1,2-Trichloroethane；*cis*-1,2-Dimethylcyclopropane；*trans*-1,2-Dimethylcyclopropane；Limonene；Camphor；Citral；1,3-Dioxolane；Tetrahydrofuran；Piperidine；2-Methyl-2-butene；3,3-Dimethyl-1-butene；Styrene；Cyclohexene；1,3-Butadiene；1,4-Cyclohexadiene；Butanone；Diethyl ketone；Isobutyric acid；Acrylonitrile；Acetylsalicylic acid；(-)-pinene；Lycopene；β-Carotene；Cholesterol；Estrone；Quinine；Strychnine；Ascorbic acid；Sucrose；[2.2]Paracyclophane；Adamantane

A.7 材料 XRD 特征峰位置的预测测试集

Cu；Au；Ni；NaCl；MgO；TiO2；CaCO3；ZrO2；CuAu；GdFeO3；ZIF-8；UiO-66；MOF-5

A.8 晶体学信息文件（CIF）生成测试集

Cu；K；NaCl；GaN；CaCO3；Benzene；Urea；Benzoic Acid；Aspirin；Ferrocene

B 本文所用全部提示词内容

B.1 分子名称与 SMILES 代码的双向转换

提示词 1

```
What is the SMILES code for {molecule}? Only output the SMILES string, with
    no additional text or explanation.
```

提示词 2

```
{
  "task_type": "name_to_smiles",
  "input_data": {
    "molecule_name": "{molecule}"
  },
  "output_format": "Return only the SMILES string as plain text, with no
    other content."
}
```

提示词 3

```
{
  "role": "You are an expert chemist.",
  "task": "Provide the SMILES string for the given chemical name. Your
    response must contain only the SMILES string itself, without any
    additional text or explanation.",
  "molecule_name": "{molecule}"
}
```

提示词 4

```
{
  "role": "You are an expert chemist specializing in chemical informatics.
    Your task is to provide the accurate ca-nonical SMILES string.",
  "instruction": "The system prompt contains examples. Follow them precisely.
     Your output must be ONLY the SMILES string.",
  "query": {
    "name": "{molecule}"
  }
}
```

提示词 5

```
What is the name of the molecule for the SMILES code {smiles_code}? Only
    output the molecule name, with no ad-ditional text or explanation.
```

提示词 6

```
{
  "task_type": "smiles_to_name",
  "input_data": {
    "smiles_code": "{smiles_code}"
```

```
  },
  "output_format": "Return only the molecule name as plain text, with no
    other content."
}
```

提示词 7

```
{
  "role": "You are an expert chemist.",
  "task": "Provide the common or IUPAC name for the given SMILES string. Your
     response must contain only the molecule name itself, without any
    additional text or explanation.",
  "smiles_code": "{smiles_code}"
}
```

提示词 8

```
{
  "role": "You are an expert chemist specializing in chemical informatics.
    Your task is to provide the common or IUPAC name for a given SMILES
    string.",
  "instruction": "The system prompt contains examples. Follow them precisely.
     Your output must be ONLY the molecule name.",
  "query": {
    "smiles": "{smiles_code}"
  }
}
```

B.2 辛醇-水分配系数（log P）预测

提示词 1

```
What is the logP Value of {molecule}? Only output logP Value, with no
    additional text or explanation.
```

提示词 2

```
{
  "task_type": "Determine the logP Value",
  "input_data": {
    "molecule_name": "{molecule}"
  },
  "output_format": "Return only the molecular logP Value as plain text, with
    no other content."
}
```

提示词 3

```
{
  "role": "You are an expert chemist specializing in perdict logP value",
  "task": "Provide the logP value of the molecule for the given chemical name
```

```
  . Your response must contain only the logP value itself, without any
  additional text or explanation.",
  "molecule_name": "{molecule}"
}
```

B.3 极性表面积（PSA）预测

提示词 1

```
What is the Polar Surface Area value of {molecule}? Only output Polar Surface
    Area Value, with no additional text or explanation.
```

提示词 2

```
{
  "task_type": "Determine the Polar Surface Area",
  "input_data": {
    "molecule_name": "{molecule}"
  },
  "output_format": "Return only the molecular Polar Surface Area Value as
   plain text, with no other content."
}
```

提示词 3

```
{
  "role": "You are an expert chemist specializing in perdict Polar Surface
   Area",
  "task": "Provide the Polar Surface Area value of the molecule for the given
     chemical name. Your response must contain only the Polar Surface Area
   value itself, without any additional text or explanation.",
  "molecule_name": "{molecule}"
}
```

B.4 分子几何构型的判定

提示词 1

```
What is the Molecular geometry of {molecule} according to VSEPR theory? Only
   output the Molecular geometry name, with no additional text or
   explanation.
```

提示词 2

```
{
  "task_type": "Determine the molecular geometry based on the VSEPR theory",
  "input_data": {
    "molecule_name": "{molecule}"
  },
```

```
  "output_format": "Return only the molecular geometry name as plain text,
   with no other content."
}
```

提示词 3

```
{
  "role": "You are an expert chemist specializing in Molecular geometry and
   VSEPR theory",
  "task": "Provide the Molecular geometry name for the given chemical name.
   Your response must contain only the Molecular geometry name itself,
   without any additional text or explanation.",
  "molecule_name": "{molecule}"
}
```

提示词 4

```
{
  "role": "You are an expert chemist specializing in Molecular geometry and
   VSEPR theory. Your task is to provide the accurate canonical Molecular
   geometry name.",
  "instruction": "The system prompt contains examples. Follow them precisely.
     Your output must be ONLY the Molecular geometry name.",
  "query": {
    "name": "{molecule}"
  }
}
```

B.5 分子点群的判定

提示词 1

```
What is the point group of {molecule}? Only output the point group name, with
     no additional text or explanation.
```

提示词 2

```
{"task_type": "Identify the molecular point group name",
  "input_data": {"molecule_name": "{molecule}"},
  "output_format": "Return only the point group name as plain text, with no
   other content.}
```

提示词 3

```
{
  "role": "You are an expert chemist specializing in structural chemistry",
  "task": "Provide the point group name for the given chemical name. Your
   response must contain only the point group name itself, without any
   additional text or explanation.",
  "molecule_name": "{molecule}"
}
```

提示词 4

```
{
  "role": "You are an expert chemist specializing in structural chemistry.
   Your task is to provide the accurate canonical point group name.",
  "instruction": "The system prompt contains examples. Follow them precisely.
    Your output must be ONLY the point group name.",
  "query": {
    "name": "{molecule}"
  }
}
```

B.6 ^{1}H NMR 谱图信号数量的预测

提示词 1

```
How many groups of peaks will {molecule} exhibit in the 1H NMR spectrum? Only
    output the number of peak groups, with no additional text or explanation
    .
```

提示词 2

```
{
  "task_type": "Determine the number of peak groups in 1H NMR spectrum",
  "input_data": {
    "molecule_name": "{molecule}"
  },
  "output_format": "Return only the number of peak groups as plain text, with
    no other content."
}
```

提示词 3

```
{
  "role": "You are an expert chemist specializing in NMR spectroscopy",
  "task": "Provide the number of peak groups of the molecule for the given
   chemical name. Your response must contain only the number of peak groups
   value itself, without any additional text or explanation.",
  "molecule_name": "{molecule}"
}
```

提示词 4

```
{
  "role": "You are an expert chemist specializing in NMR spectroscopy. Your
   task is to provide the number of peak groups in 1H NMR spectrum for a
   given molecule.",
  "instruction": "The system prompt contains examples. Follow them precisely.
    Your output must be ONLY the number of peak groups.",
  "query": {
    "molecule_name": "{molecule}"
  }
```

```
}
```

B.7 XRD 谱图特征峰位置的预测

提示词 1

```
How many peaks will a well-crystallized {molecule} material exhibit in the
    XRD pattern using Cu αK radiation source (test range 10°-80°)? Output
    only the positions of the peaks, without any additional text or
    explanation.
```

提示词 2

```
{
  "task_type": "Determine the positions of peaks in XRD spectrum (test range
    10°-80°) using a Cu αK radiation source (Assume that the material has
    good crystallinity)",
  "input_data": {
    "molecule_name": "{molecule}"
  },
  "output_format": "Return only the positions of the peaks as plain text,
    with no other content."
}
```

提示词 3

```
{
  "role": "You are an expert chemist specializing in X-ray diffraction",
  "task": "Provide the positions of peaks for the given material (test range
    10°-80°). Your response must contain only the peaks itself, without any
    additional text or explanation.",
  "molecule_name": "{molecule}"
}
```

提示词 4

```
{
  "role": "You are an expert chemist specializing in X-ray diffraction. Your
    task is to provide the positions of the peaks for a given material (test
    range 10°-80°).",
  "instruction": "The system prompt contains examples. Follow them precisely.
     Your output must be ONLY the positions of the peaks.",
  "query": {
    "molecule_name": "{molecule}"
  }
}
```

B.8 晶体学信息文件（CIF）的模拟与生成测试

提示词 1

```
Generate the cif file of the {molecule}, and output only the complete content
    of the cif file without additional text or explanation.
```

提示词 2

```
{
  "task_type": "Generate cif file",
  "input_data": {
    "molecule_name": "{molecule}"
  },
  "output_format": "Return only the complete content of the cif file, with no
    other content."
}
```

提示词 3

```
{
  "role": "You are an expert chemist specializing in generate cif file",
  "task": "the complete content of the cif file for the given material or
   molecule. Your response must contain only the complete content of the cif
    file, without any additional text or explanation.",
  "molecule_name": "{molecule}"
}
```

提示词 4

```
{
  "role": "You are an expert chemist specializing in generate cif file. Your
   task is to provide the complete con-tent of the cif file for a given
   material or explanation.",
  "instruction": "The system prompt contains examples. Follow them precisely.
    Your output must be ONLY the complete content of the cif file, without
   any additional text or explanation.",
  "query": {
    "molecule_name": "{molecule}"
  }
}
```